Junbin Huang

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Personal

Citizen of China
Also known as: Jun-Bin Huang
Chinese name: 黄峻彬

Research Interests

Quantitative Finance and Machine Learning

AI4Math

Finite Element Methods, and Numerical Methods in general

Selected Experience

2023–June 2024 **Quantitative Researcher** at Starvast Quant, focusing primarily on deep learning methods for identifying investment opportunities.

Dec. 2020–Dec. 2022 **Boya Postdoctoral Fellow** at Beijing International Center for Mathematical Research, Peking University. My supervisor was Prof. Bin Dong. This appointment was funded by the International Postdoctoral Exchange Fellowship Program (Talent-Introduction Program). I studied machine learning approaches for solving differential equations and the underlying mathematics.

Feb. 2020–May 2020 **Teaching Assistant** for Prof. Wim van Rees, in MIT Subject 2.081 *Plates & Shells* (Graduate level).

Sept. 2017–May 2020 **Research Assistant** for Prof. Klaus-Jürgen Bathe, in the Department of Mechanical Engineering, Massachusetts Institute of Technology. The main job was developing overlapping finite elements with applications.

Education

May 2020 **Ph.D.** in Mechanical Engineering and Computation, Massachusetts Institute of Technology
Thesis Title: New Overlapping Finite Elements and Their Application in the AMORE Paradigm (Advisor: Professor Klaus-Jürgen Bathe)
GPA: 5.0/5.0

June 2017 **M.Eng.** with outstanding master's thesis, in Mechanics, Tsinghua University
Thesis Title: Developments of High-Performance Finite Element Models Based on the Analytical Trial Functions (Advisor: Professor Song Cen)
GPA: 92.5/100 (ranking 1/51)

July 2014 **B.Eng.** with honors, in Civil Engineering, Tsinghua University
GPA: 90/100 (ranking 7/92)

July 2014 **B.Sc.** in Pure and Applied Mathematics (Second Bachelor's Degree), Tsinghua University

Service

Reviewer: Computers & Structures, Engineering Computations, Journal of Computational Physics, Computer Methods in Applied Mechanics and Engineering

Programming Skills

Python, Linux, MATLAB, C++, Fortran, LaTeX

Selected Publications

2021 1. **J Huang**, KJ Bathe*. On the convergence of overlapping elements and overlapping meshes. *Computers & Structures*, 2021, 244: 106429.

2020 2. **J Huang**, KJ Bathe*. Overlapping finite element meshes in AMORE. *Advances in Engineering Software*, 2020, 144: 102791.

2019 3. **J Huang**, KJ Bathe*. Quadrilateral overlapping elements and their use in the AMORE paradigm. *Computers & Structures*, 2019, 222: 25–35.

2018 4. **J Huang**, S Cen*, Z Li, CF Li. An unsymmetric 8-node hexahedral solid-shell element with high distortion tolerance: Linear formulations. *International Journal for Numerical Methods in Engineering*, 2018, 116(12–13): 759–783.